

Thermochemical Non-equilibrium Modeling of an N_2/N Shock Layer Using a Two-Temperature Formulation

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Abstract

High-speed flows produce strong shock waves that create extreme thermodynamic conditions, resulting in thermochemical non-equilibrium. A numerical simulation of the jump across a shock is used to model the non-equilibrium flow in nitrogen, utilizing a two-temperature formulation. In this simulation, the Rankine–Hugoniot relations are used to model the shock as a frozen discontinuity, and the gas is treated as a mixture of molecular nitrogen (N_2) and atomic nitrogen (N). The chemical dissociation and vibrational relaxation that describe the downstream equilibrium flow are formulated using a system of first-order ordinary differential equations. Results show an instantaneous jump in translational temperature behind the shock, followed by gradual relaxation of vibrational and chemical modes downstream until equilibrium is reached. Overall, the simulation captures the expected relaxation behavior of non-equilibrium nitrogen flow behind strong shock waves.

1 Introduction

Hypersonic flow creates strong shock waves that, when jumped across, rapidly convert kinetic energy into internal energy. This process occurs almost instantaneously and produces a sharp increase in post-shock temperatures. Translational energy responds immediately, while vibrational excitation and chemical reactions occur over a finite time scale. This difference in response is what leads to thermochemical nonequilibrium.

In this project, the thermochemical nonequilibrium flow in nitrogen behind a strong normal shock wave is modeled numerically. The gas is treated as a mixture of molecular nitrogen (N_2) and atomic nitrogen (N) using a two-temperature formulation to capture nonequilibrium between translational–rotational and vibrational–electronic energy modes. The flow is modeled as steady and one-dimensional, with the shock treated as a discontinuity in the flow properties.

The solution is organized into three regions. The pre-shock region represents the known upstream equilibrium state and provides the initial conditions. At the shock, the Rankine–Hugoniot relations are applied to enforce conservation of mass, momentum, and energy, resulting in a frozen post-shock state where only translational energy responds instantaneously. Downstream of the shock, the governing equations are reduced to a canonical form and integrated to model vibrational relaxation and chemical dissociation. This approach allows the flow to relax gradually toward equilibrium, capturing the key physical behavior expected in non-equilibrium nitrogen flow behind strong shock waves.

2 Key Assumptions

- The gas mixture is modeled as a mixture of perfect gases.

- Vibrational energy is modeled separately using a two-temperature approximation.
- The flow is inviscid and governed by the species mass, momentum, energy, and vibrational/electronic conservation laws.
- The system can be approximated as quasi-steady immediately behind the shock.

3 Governing Equations

The governing equations for a steady, inviscid, one-dimensional flow of a reacting gas mixture are written in conservative form,

$$\frac{d}{dx} \begin{pmatrix} \rho_s u \\ p + \rho u^2 \\ \rho u H \\ \rho u e_v \end{pmatrix} = \begin{pmatrix} \omega_s \\ 0 \\ 0 \\ \Omega_v \end{pmatrix}. \quad (1)$$

In order to numerically integrate the governing equations, they are reduced to a canonical system of ordinary differential equations. The primitive variable vector and its spatial derivatives are defined accordingly,

$$\mathbf{y} = \begin{pmatrix} Y_s \\ u \\ T \\ T_v \end{pmatrix}, \quad \frac{d\mathbf{y}}{dx} = \begin{pmatrix} s_{Y_s} \\ s_u \\ s_T \\ s_{T_v} \end{pmatrix}. \quad (2)$$

The source terms are derived using the constant mass flux assumption, $\dot{m} = \rho u$, where the source terms are denoted by $\mathbf{s}$.

3.1 Species Conservation

The species conservation equation begins from the steady one-dimensional continuity equation

$$\frac{d}{dx}(\rho u Y_s) = \omega_s,$$

where ω_s is the mass production rate of species s . Using the constant mass flux assumption $\dot{m} = \rho u$, this reduces directly to the canonical form

$$\frac{dY_s}{dx} = \frac{\omega_s}{\dot{m}}.$$

3.2 Momentum Equation

The momentum equation is written in inviscid steady form as

$$\frac{d}{dx}(p + \rho u^2) = 0.$$

After careful differentiation, a linear relationship between the velocity and temperature gradients is obtained, which is coupled with the energy equation. The canonical vibrational temperature equation is obtained by expressing the vibrational energy in terms of the vibrational temperature, accounting for chemical energy exchange.

3.3 Translational Energy Equation

The total energy equation is expressed in terms of the mixture total enthalpy,

$$H = h + \frac{u^2}{2},$$

and expanded using constant mass flux. Substituting the species source terms produces a second linear equation coupling $\frac{du}{dx}$ and $\frac{dT}{dx}$, which together with the momentum equation forms a 2×2 linear system.

Solving this system using Cramer's rule yields the explicit canonical expressions

$$\begin{aligned} \frac{du}{dx} &= \frac{-\frac{uRTc_p}{\dot{m}} \sum_i \frac{\omega_i}{M_i} + \frac{Ru}{\dot{m}M_{\text{mix}}} \sum_i h_i(T)\omega_i}{\det(A)}, \\ \frac{dT}{dx} &= \frac{-\frac{1}{\dot{m}} \left(u^2 - \frac{RT}{M_{\text{mix}}} \right) \sum_i h_i(T)\omega_i + \frac{u^2 RT}{\dot{m}} \sum_i \frac{\omega_i}{M_i}}{\det(A)}, \end{aligned}$$

with

$$\det(A) = u^2 c_p - \frac{R}{M_{\text{mix}}} (T c_p + u^2).$$

3.4 Vibrational Energy Equation

The vibrational energy equation is obtained from

$$\frac{d}{dx}(\rho u e_v) = \Omega_v,$$

where e_v is the mixture vibrational energy per unit mass.

The canonical vibrational temperature equation is obtained by expressing e_v in terms of the vibrational temperature. This accounts for the chemical energy exchange,

$$\frac{dT_v}{dx} = \frac{1}{\dot{m} c_{v,v}} \left(\Omega_v - \sum_m e_v^{(m)} \omega_m \right).$$

3.5 Final Canonical ODE System

Collecting the source terms, the governing equations are reduced to a first-order canonical system for the primitive variable vector,

$$\mathbf{y} = \begin{pmatrix} Y_s \\ u \\ T \\ T_v \end{pmatrix}.$$

Below is the final set of ODEs that are used to numerically compute the relaxation downstream,

$$\frac{d\mathbf{y}}{dx} = \begin{pmatrix} \frac{dY_s}{dx} \\ \frac{du}{dx} \\ \frac{dT}{dx} \\ \frac{dT_v}{dx} \end{pmatrix} = \begin{pmatrix} \frac{\omega_s}{\dot{m}} \\ \frac{1}{\det(A)} \left(-\frac{uRTc_p}{\dot{m}} \sum_i \frac{\omega_i}{M_i} + \frac{Ru}{\dot{m}M_{\text{mix}}} \sum_i h_i(T)\omega_i \right) \\ \frac{1}{\det(A)} \left(-\frac{1}{\dot{m}} \left(u^2 - \frac{RT}{M_{\text{mix}}} \right) \sum_i h_i(T)\omega_i + \frac{u^2 RT}{\dot{m}} \sum_i \frac{\omega_i}{M_i} \right) \\ \frac{1}{\dot{m} c_{v,v}} \left(\Omega_v - \sum_m e_v^{(m)} \omega_m \right) \end{pmatrix}.$$

This canonical system constitutes the final set of coupled ordinary differential equations integrated numerically to compute the downstream relaxation of the flow.

4 Physical Models

4.1 Vibrational–Translational Energy Exchange

Vibrational-translational energy exchange is modeled using the Millikan-White vibrational relaxation formulation. The vibrational relaxation time, τ_v , is often expressed in terms of pressure and temperature through empirical relations

$$\tau_v(T, p) \propto \left(\frac{1}{p}\right) \exp \left[a(T^{-1/3}) - b \right] - 18.42,$$

with fixed coefficients a and b appropriate for N_2 , incorporating pressure dependence and the characteristic $T^{-1/3}$ temperature scaling.

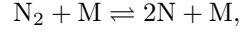
The resulting relaxation time is used directly to define the vibrational energy source term,

$$\Omega_v = \frac{\rho (e_v^*(T) - e_v(T_v))}{\tau_v},$$

which governs the evolution of the vibrational temperature downstream of the shock.

4.2 Nitrogen Dissociation Kinetics

Chemical nonequilibrium is modeled using the single reversible reaction



where M is either N_2 or N . The forward dissociation rate is written in Arrhenius form, with the controlling temperature T_x taken as

$$T_x = \sqrt{TT_v},$$

following Park’s vibrational nonequilibrium model.

4.2.1 Backward Reaction and Equilibrium Constant

The backward recombination rate is obtained using detailed balance,

$$k_b(T) = \frac{k_f(T_x)}{K_c(T)},$$

where the equilibrium constant for nitrogen is approximated as

$$K_c(T) = 18 \exp \left(-\frac{113,000}{T} \right) \text{ mol/cm}^3.$$

5 Numerical Methods

5.1 Solution of the Rankine-Hugoniot Jump Relations

The frozen post-shock state is obtained by solving the one-dimensional Rankine–Hugoniot jump relations, which enforce conservation of mass, momentum, and energy across a normal shock. These relations are used to compute the post-shock thermodynamic state that serves as the initial condition for the downstream nonequilibrium relaxation problem.

5.1.1 Pre-shock State

The upstream flow is fully specified by the pre-shock pressure P_1 , temperature T_1 , shock speed U_s , and mixture composition. The pre-shock density is computed assuming ideal-gas behavior,

$$\rho_1 = \frac{M_{\text{mix}} P_1}{RT_1},$$

where M_{mix} is the mixture molar mass and R is the universal gas constant.

5.1.2 Frozen-Flow Assumption

Across the shock, chemistry and vibrational energy are assumed to be frozen. As a result, the species mass fractions and vibrational temperature remain unchanged through the discontinuity,

$$Y_{s,2} = Y_{s,1}, \quad T_{v,2} = T_{v,1}.$$

Only the translational temperature and flow velocity are allowed to change across the shock.

5.1.3 Continuity and Momentum Relations

The post-shock density and velocity are related through conservation of mass,

$$\rho_2 U_2 = \rho_1 U_s, \quad U_2 = \frac{\rho_1}{\rho_2} U_s.$$

Conservation of momentum provides an expression for the post-shock pressure,

$$P_2 = P_1 + \rho_1 U_s^2 \left(1 - \frac{\rho_1}{\rho_2} \right).$$

5.1.4 Energy Equation and Iterative Solution

The post-shock temperature is determined from conservation of total energy. Writing the energy balance in terms of the mixture enthalpy,

$$H_1 + \frac{1}{2} U_s^2 = H_2 + \frac{1}{2} U_2^2,$$

the post-shock temperature T_2 is obtained by iteratively solving the energy equation. A Newton iteration is employed, updating the temperature according to

$$T_2^{(k+1)} = T_2^{(k)} - \frac{H_2 - H_1 - \frac{1}{2} U_s^2 \left(1 - \left(\frac{\rho_1}{\rho_2} \right)^2 \right)}{c_{p,2}},$$

where $c_{p,2}$ is the post-shock specific heat at constant pressure. Iteration continues until convergence of the post-shock temperature is achieved.

The resulting frozen post-shock state $(\rho_2, U_2, P_2, T_2, T_{v,2}, Y_{s,2})$ is then used to initialize the downstream nonequilibrium ODE solver.

5.1.5 Convergence Criteria for the Rankine-Hugoniot Solution

Convergence of the Rankine-Hugoniot solution is assessed by monitoring the residuals associated with conservation of mass, momentum, and energy across the shock. The mass conservation residual is defined as

$$R_1 = \frac{\rho_1 U_s - \rho_2 U_2}{\rho_1 U_s},$$

which measures the mismatch in mass flux across the shock. The momentum conservation residual is given by

$$R_2 = \frac{P_1 + \rho_1 U_s^2 - (P_2 + \rho_2 U_2^2)}{P_1 + \rho_1 U_s^2},$$

and quantifies the error in momentum balance. The energy conservation residual is written as

$$R_3 = \frac{H_1 + \frac{1}{2}U_s^2 - (H_2 + \frac{1}{2}U_2^2)}{H_1 + \frac{1}{2}U_s^2},$$

representing the deviation from total energy conservation. The iterative solution procedure is considered converged when the absolute values of all residuals fall below a prescribed tolerance,

$$|R_1|, |R_2|, |R_3| < 10^{-14},$$

ensuring that the frozen post-shock state satisfies the Rankine-Hugoniot relations to machine precision.

5.2 Computational Steps Summary

1. **Define upstream state (region 1).** Set the pre-shock conditions $(P_1, T_1, T_{v,1}, U_s)$ and upstream composition (pure N_2), and compute ρ_1 from the ideal gas law using R_{N_2} .
2. **Compute frozen post-shock state (initial condition at $x = 0^+$).** Call a Rankine-Hugoniot solver that assumes **frozen chemistry** and **frozen vibration** across the discontinuity, so $Y_{s,2} = Y_{s,1}$ and $T_{v,2} = T_{v,1}$. The RH solver iterates using momentum to update P_2 and an energy residual to update T_2 until normalized residuals R_1, R_2, R_3 fall below tolerance.
3. **Set constant mass flux and initial ODE state.** After RH, compute ρ_2 and define the constant mass flux $G = \rho_2 U_2$. The downstream ODE state is initialized as $\mathbf{y}(0^+) = [Y_{N_2,2}, U_2, T_2, T_{v,2}]^T$.
4. **Integrate the canonical ODE system downstream with ode45.** Integrate from $x = 0$ to $x_{\max} = 2.5$ m using ode45, with the RHS provided by $\text{rhs}(x, \mathbf{y}, G)$.

5.3 RHS Evaluation Sequence at Each Spatial Location

At each integration point x , the RHS routine `rhs` follows this sequence:

5. **Reconstruct thermodynamic state from (Y_{N_2}, u, T, T_v) .** Compute mixture molecular weight and gas constant from Y_{N_2} (with $Y_N = 1 - Y_{N_2}$), then compute $\rho = G/u$ (from constant mass flux) and $p = \rho R_{\text{mix}} T$ (ideal gas mixture).
6. **Finite-rate chemistry.** Evaluate the net production rate for the single reversible reaction $N_2 + M \rightleftharpoons 2N + M$. The forward dissociation uses Park's controlling temperature $T_x = \sqrt{TT_v}$. The backward rate is obtained via detailed balance using the equilibrium constant $K_c(T)$. This defines the species source term ω_{N_2} .
7. **Species equation.** Advance the composition using the canonical species equation $\frac{dY_{N_2}}{dx} = \frac{\omega_{N_2}}{G}$.
8. **Coupled momentum-energy update.** Compute $\frac{du}{dx}$ and $\frac{dT}{dx}$ from the coupled momentum/energy subsystem using the explicit determinant-based (Cramer's rule) form of the canonical equations, including the chemical enthalpy coupling through $\sum_i h_i(T)\omega_i$.
9. **VT relaxation.** Compute the vibrational energy relaxation source Ω_v using the Millikan-White relaxation time $\tau_v(T, p)$, then advance T_v using the canonical vibrational temperature equation $\frac{dT_v}{dx} = \frac{\Omega_v}{\rho u Y_{N_2} c_{v,v}}$.

5.4 Post-processing and Outputs

After integration, the solver output is post-processed:

- Compute $Y_N = 1 - Y_{N_2}$, M_{mix} , and R_{mix} .
- Reconstruct $\rho = G/u$ and $p = \rho R_{\text{mix}} T$.
- Auto-detect an “equilibrium” location x_{eq} by finding the first point where the key relaxing variables (T, T_v, Y_{N_2}) are within a prescribed relative tolerance of their end-of-domain values.
- Print 4 state blocks (pre-shock, frozen post-shock, detected equilibrium, and $x = x_{\text{max}}$).
- Generate full-domain and zoomed plots, plus internal distributions at specified x .

6 Results

The computed flow variables at four distinct physical locations—the pre-shock state, the frozen post-shock state (initial condition), the detected equilibrium location, and the final downstream boundary—are summarized in Table 1. The equilibrium position was numerically detected at $x \approx 1.86$ m.

Table 1: Summary of Flow Variables at Key Relaxation Stages

Location	Symbol	Pre-Shock ($x < 0$)	Post-Shock ($x = 0^+$)	Equilibrium ($x \approx 1.86$ m)	Downstream ($x = 2.5$ m)
Position (m)	x	—	0.000	1.86	2.50
Translational Temp. (K)	T	200	23118	6530	6530
Vibrational Temp. (K)	T_v	200	200	6530	6530
Velocity (m/s)	U	7000	1177	543	543
Pressure (kPa)	P	0.005	3.44	3.81	3.81
Density ($\rho \times 10^4$, kg/m ³)	ρ	0.842	5.08	11.2	11.2
N ₂ Mass Fraction	Y_{N_2}	1.000	1.000	0.189	0.189
N Mass Fraction	Y_N	0.000	0.000	0.811	0.811

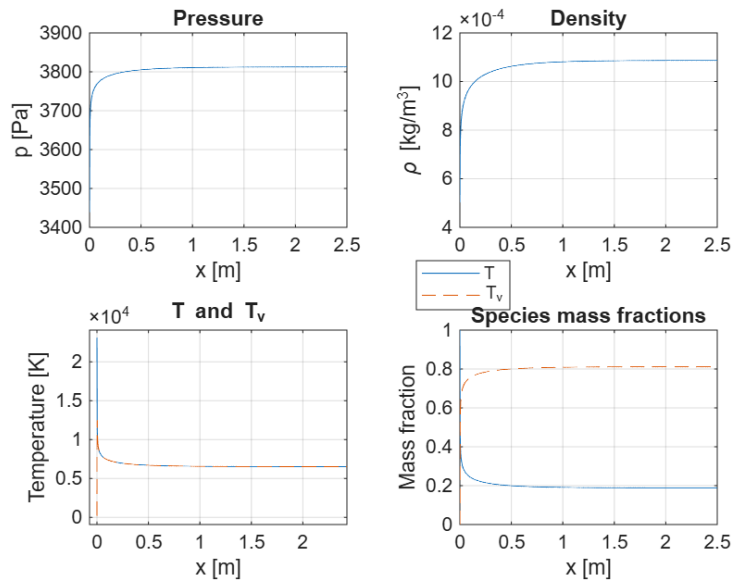


Figure 1: Full domain downstream, $x = 2.5$ m.

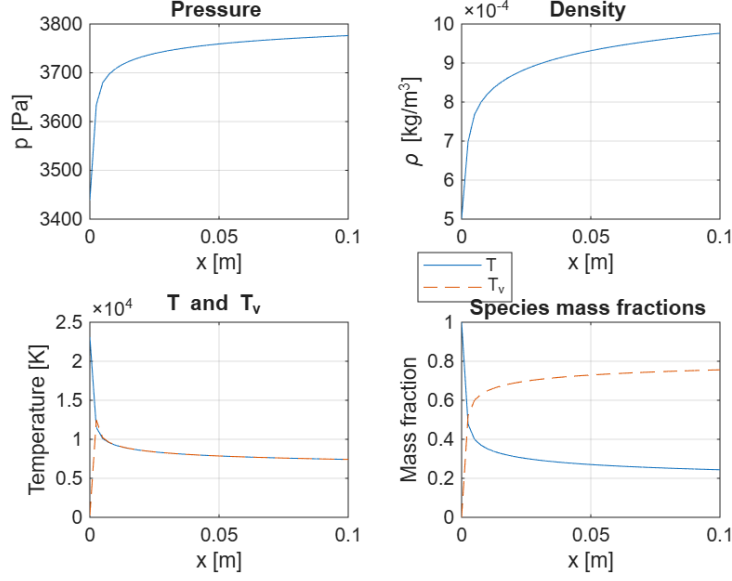


Figure 2: Zoomed domain, $x = 0.1$ m.

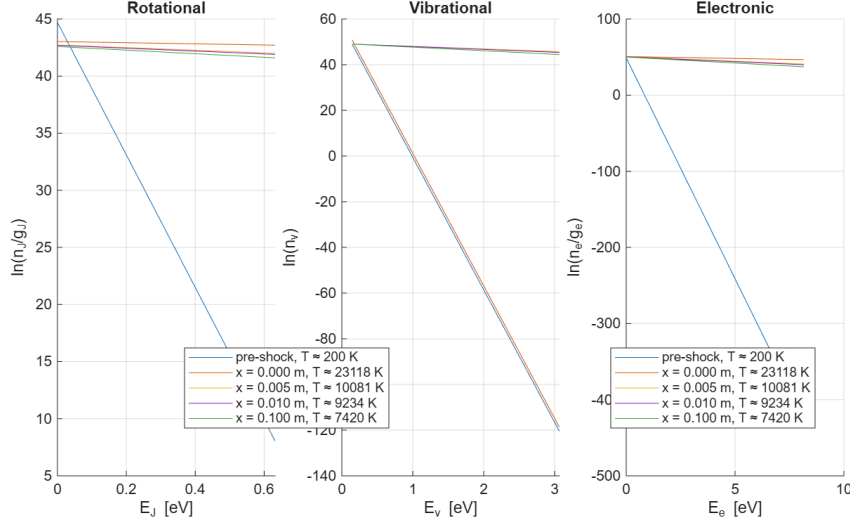


Figure 3: Internal distribution.

7 Discussion

The simulation accurately models the thermochemical nonequilibrium relaxation of a strong normal shock in a nitrogen mixture. The pre-shock parameters serve as the initial conditions for the system of ordinary differential equations that define the jump across the shock and the downstream relaxation region. The initial conditions are given by $P_1 = 5$ Pa, $\rho_1 = 8.42 \times 10^{-5}$ kg/m³, $T_1 = T_{v,1} = 200$ K, and $U_1 = 7000$ m/s.

After applying the Rankine–Hugoniot relations, the frozen-flow assumption is confirmed by the instantaneous exchange between kinetic and internal energy. The velocity drops to $U_2 = 1177$ m/s, while the translational temperature rises sharply to $T_2 = 23118$ K. Immediately behind the shock, the vibrational temperature remains unchanged at $T_{v,2} = 200$ K, indicating that vibrational energy has not yet relaxed. This behavior is further confirmed by the species composition, which remains identical to the pre-shock state with $Y_{N_2} = 1$ and $Y_N = 0$.

Figure 2 shows the near-shock region ($x > 0$), where the translational temperature decreases rapidly from its frozen value of approximately 23,000 K to below 10,000 K within a few centimeters. This decrease

corresponds to energy redistribution into internal modes. Simultaneously, the vibrational temperature rises sharply from 200 K to approximately 13,000 K due to strong vibrational–translational coupling. Dissociation is observed concurrently, confirming that vibrational excitation and chemical dissociation act as energy sinks for the translational mode.

Further downstream, the flow approaches equilibrium. At $x = 2.5$ m, the translational and vibrational temperatures converge to approximately $T = T_v = 6530$ K. The species mass fractions approach equilibrium values of $Y_{\text{N}_2} \approx 0.189$ and $Y_{\text{N}} \approx 0.811$. The velocity stabilizes near 543 m/s, and the pressure increases modestly from 3.44 kPa to approximately 3.81 kPa. The equilibrium location is detected numerically at $x \approx 1.86$ m. Overall, the results follow the expected physical trends for nonequilibrium nitrogen flow behind a strong shock.

8 Conclusion

A one-dimensional steady model was developed to simulate thermochemical nonequilibrium relaxation downstream of a strong normal shock in a nitrogen mixture consisting of N_2 and N . The model was organized into three regions, with pre-shock conditions providing the initial state and the Rankine–Hugoniot relations used to compute the frozen post-shock conditions. These conditions were used to initialize a coupled system of ordinary differential equations derived from the governing equations.

The simulation accurately captures the downstream vibrational relaxation and chemical dissociation processes. The translational temperature increases sharply across the shock while the velocity decreases, demonstrating the exchange between kinetic and internal energy. Immediately downstream, vibrational temperature rises as translational energy decreases, and the flow eventually reaches thermochemical equilibrium. Analysis of macroscopic flow properties and internal energy distributions confirms the correct ordering of relaxation processes and highlights the role of vibrational excitation and chemical dissociation as dominant energy sinks. Overall, the simulation provides a numerically reliable and physically consistent description of shock structure and energy transfer mechanisms in high-temperature nitrogen flows.